Introduction

In this project, I analyze user behavior for Cyclistic, a fictional bike-share company based in Chicago. The company offers bikes that can be used through two types of subscriptions: casual riders, who purchase single rides or day passes, and annual members, who pay for a yearly subscription.

The purpose of this analysis is to explore how these two types of users interact differently with the service. Understanding these behavioural differences is important because annual members represent a more stable and profitable customer segment for the company.

Using historical trip data, I explore patterns such as ride frequency, average ride duration, and usage by day of the week. The analysis focuses on identifying trends that can help explain how casual riders use the service compared to members and what opportunities may exist to encourage casual riders to become long-term subscribers.

Business Task

The business objective of this project is to understand the key differences between casual riders and annual members in their bike usage habits.

By analyzing ride data and user behavior, the goal is to generate insights that could help Cyclistic's marketing team design strategies aimed at converting casual riders into annual members.

The main question guiding this analysis is:

How do casual riders and annual members use Cyclistic bikes differently, and how can these insights support marketing decisions to increase memberships?

Data Sources

The data used in this project comes from Cyclistic's historical trip data, publicly available through the Divvy Bike Share program in Chicago. The dataset is distributed as monthly CSV files, each containing detailed information about individual bike trips.

For this analysis, I used the most recent 12 months of available data, resulting in a combined dataset of more than 5.6 million rides.

The following files were included:

- 202503-divvy-tripdata.csv

- 202504-divvy-tripdata.csv
- 202505-divvy-tripdata.csv
- 202506-divvy-tripdata.csv
- 202507-divvy-tripdata.csv
- 202508-divvy-tripdata.csv
- 202509-divvy-tripdata.csv
- 202510-divvy-tripdata.csv
- 202511-divvy-tripdata.csv
- 202512-divvy-tripdata.csv
- 202601-divvy-tripdata.csv
- 202602-divvy-tripdata.csv

Each dataset contains trip-level information including:

- ride_id
- rideable_type
- started_at
- ended_at
- start_station_name
- end_station_name
- member_casual

The data is anonymized and does not include personally identifiable information, making it suitable for behavioral analysis focused on usage trends.

Process: Data Cleaning and Transformation

After importing all monthly CSV files into Excel, Power Query was used to append the datasets into a single consolidated table. This ensured that all months shared the same structure before analysis.

Data Cleaning Steps

The following transformations were applied:

- Converted **started_at** and **ended_at** columns to DateTime format.
- Verified column consistency across all files before merging.
- Removed rows containing null or invalid values.
- Checked for negative or unrealistic ride durations.

New Columns Created

To support the analysis, several calculated columns were added:

1. Ride Length

Calculated as the difference between ride end and start time:

ride_length = ended_at - started_at

This column was later converted into minutes for easier comparison between user types.

2. Ride Length (Minutes)

ride_length_minutes = Duration.TotalMinutes([ride_length])

This allowed average ride duration to be interpreted more clearly in visualizations.

3. Day of Week

Extracted from the ride start timestamp:

Date.DayOfWeekName([started_at])

4. Weekday Number

Created to ensure correct weekday ordering in charts:

Date.DayOfWeek([started_at], Day.Monday) + 1

Handling Large Data Volume

Because the combined dataset exceeded Excel's worksheet row limit, the data was loaded into the Excel Data Model instead of a standard worksheet. This allowed analysis of all 5.6 million records using pivot tables without reducing the dataset size.

These preparation steps transformed raw trip data into a clean and structured dataset ready for analysis.

Data Validation and Exploration with SQL (BigQuery)

Because the dataset contains more than five million records, part of the analysis workflow was replicated in Google BigQuery to simulate how large datasets are commonly explored in a database environment.

SQL was used to perform exploratory checks and validate key metrics later analyzed in Excel.

Dataset Exploration

First, a general overview of the dataset size was obtained:

```
SELECT COUNT(*) AS total_rides

FROM cyclistic_trips;
```

This confirmed the total number of records included in the analysis.

Rides by User Type

To understand the distribution of riders, the following query was used:

```
SELECT

  member_casual,

  COUNT(*) AS total_rides

FROM cyclistic_trips

GROUP BY member_casual

ORDER BY total_rides DESC;
```

This helped verify that annual members generated the majority of rides, consistent with results observed in Excel.

Average Ride Duration

Ride duration was calculated directly using SQL timestamp functions:

```
SELECT

  member_casual,

  AVG(TIMESTAMP_DIFF(ended_at, started_at, MINUTE)) AS avg_ride_minutes

FROM cyclistic_trips

GROUP BY member_casual;
```

This query confirmed that casual riders tend to have longer rides compared to annual members.

Weekly Usage Patterns

To analyze behavioral differences throughout the week:

```
SELECT
  EXTRACT(DAYOFWEEK FROM started_at) AS weekday,
  member_casual,
  COUNT(*) AS rides
FROM cyclistic_trips
GROUP BY weekday, member_casual
ORDER BY weekday;
```

This supported the observation that members ride more frequently during weekdays, while casual riders show increased activity on weekends.

Using SQL alongside Excel demonstrated how the same analytical questions can be explored both in spreadsheet tools and scalable database environments.

Analysis

Overview of the Analysis Process

After cleaning and validating the Cyclistic dataset using Excel and SQL, the next step consisted of analysing user behaviour through a set of visualisations created in Excel. The objective of this analysis was to identify differences between **casual riders** and **annual members**, focusing on usage patterns, ride duration, and temporal trends.

The dashboard was designed to answer the business question:

How do casual riders and annual members use Cyclistic bikes differently?

To support this objective, several key metrics were analysed:

- Total number of rides by user type
- Average ride duration
- Ride distribution by weekday
- Behavioural patterns over time

Each visualization focuses on revealing behavioral differences that could support marketing decisions.

Ride Volume by User Type

The first chart compares the total number of rides completed by **casual users** and **members**.

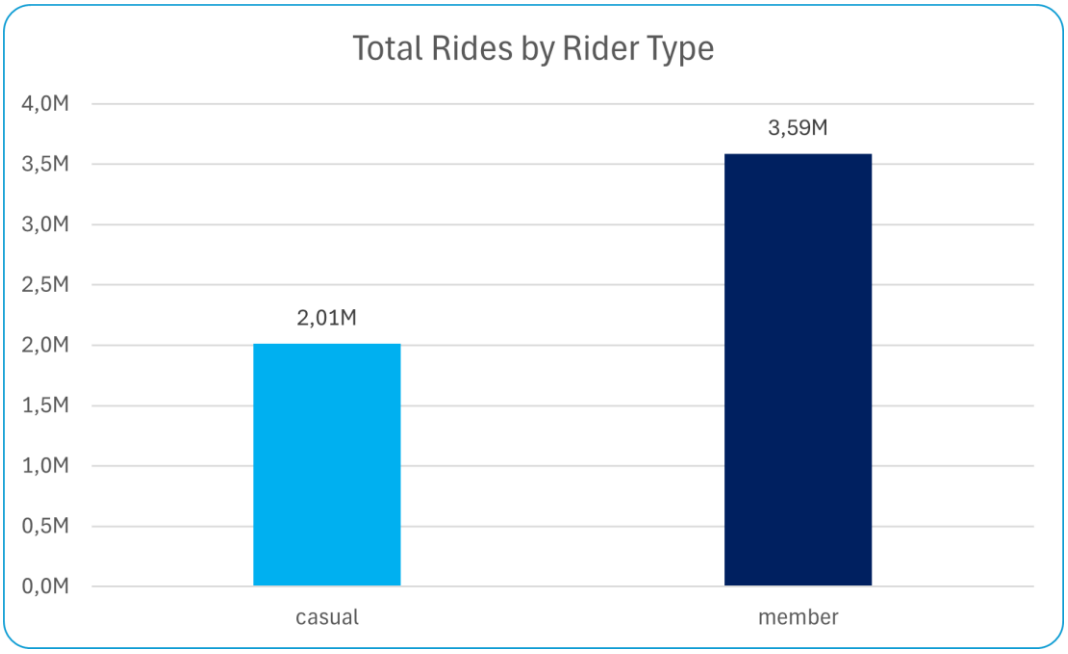

Figure 1. Total number of rides by user type.

Source: Cyclistic dataset (2026), author's analysis.

The analysis shows a clear difference in overall usage volume. Members account for a larger proportion of total rides, suggesting that subscriptions encourage frequent and consistent usage. Casual riders, while still significant, represent a smaller share of total activity.

This result aligns with expectations: annual members tend to integrate Cyclistic bikes into their daily routines, while casual users appear to rely on the service occasionally.

Interpretation:

- Membership correlates with higher ride frequency.
- Members likely use bikes as a regular transportation method rather than leisure-only activity.

Average Ride Duration

Another key metric analysed was the **average ride duration**, displayed using a chart where the time axis was converted into minutes for better readability.

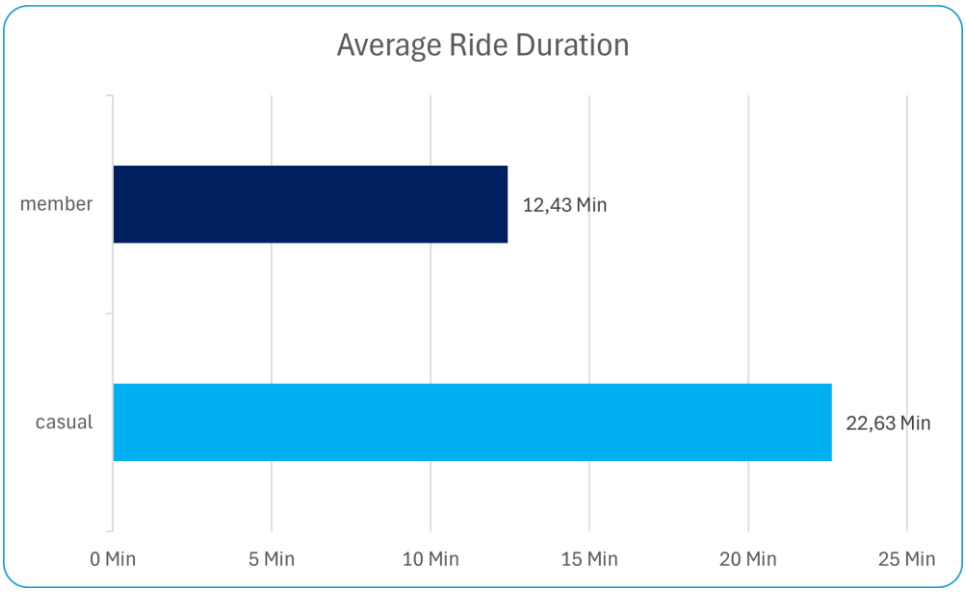

Figure 2. Average ride duration by user type (minutes)

Source: Cyclistic dataset (2026), author's analysis.

The results indicate that:

- Casual riders generally have **longer average ride durations**.
- Members tend to complete **shorter but more frequent rides**.

This behavioural contrast suggests different motivations:

- Casual riders likely use bikes for leisure, tourism, or recreational purposes.
- Members appear to use bikes for commuting or short-distance travel.

Formatting the axis into minutes helped make these differences clearer and easier to interpret visually.

Interpretation:

- Casual usage = longer, experience-oriented rides.
- Member usage = efficient, purpose-driven trips.

Ride Distribution by Day of the Week

A key improvement in the analysis involved correctly ordering weekdays using a **weekday number column**, ensuring the chart followed chronological order instead of alphabetical order.

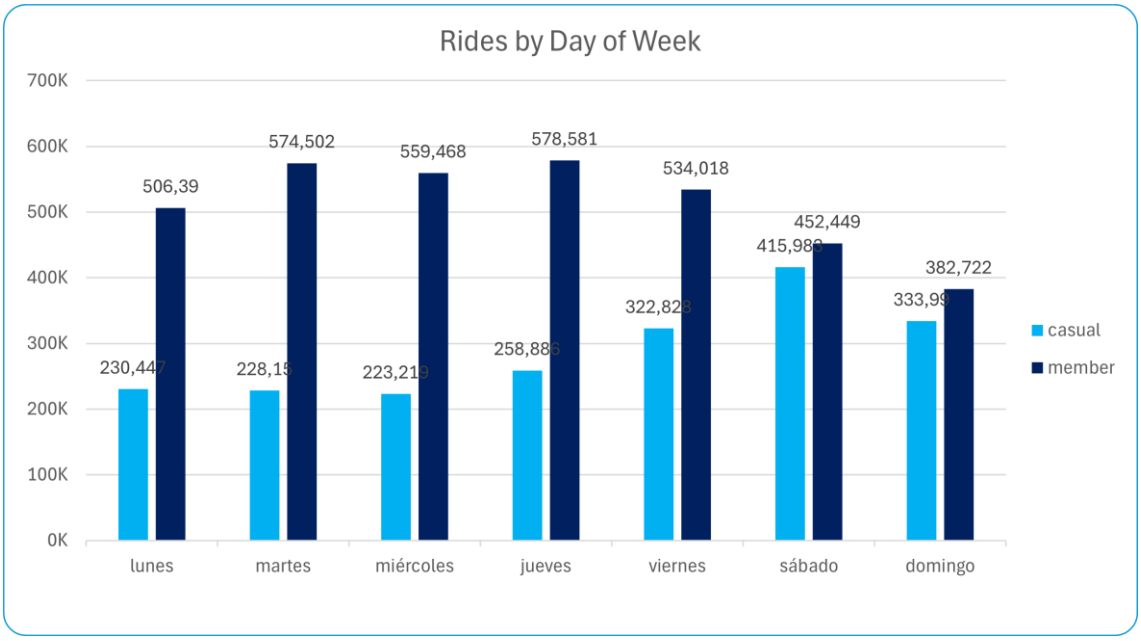

Figure 3. Number of rides by weekday and user type

Source: Cyclistic dataset (2026), author's analysis.

Once corrected, a clear weekly pattern emerged:

- Member rides remain relatively stable throughout weekdays.
- Casual rides increase noticeably during weekends.

This indicates distinct behavioural cycles:

- Members rely on bikes during working days.
- Casual riders are more active during leisure periods.

The weekday ordering was essential to accurately reveal this pattern, as incorrect sorting would have hidden the temporal trend.

Interpretation:

- Members → weekday commuting behaviour.
- Casual riders → weekend recreational behaviour.

Bike Type Usage

In addition to analysing ride frequency and duration, bike type usage was explored to better understand differences in rider preferences between casual users and annual members.

The results show that classic bikes represent the majority of rides for both user types, suggesting they remain the standard option for most trips. However, casual riders display a slightly higher proportion of electric bike usage compared to members.

This pattern may indicate that casual riders prioritize convenience and comfort, especially for longer or recreational rides, while members tend to rely on bicycles as part of their regular routines and commuting habits.

Understanding these preferences can help Cyclistic adapt both bike availability and marketing strategies depending on rider behaviour and usage context.

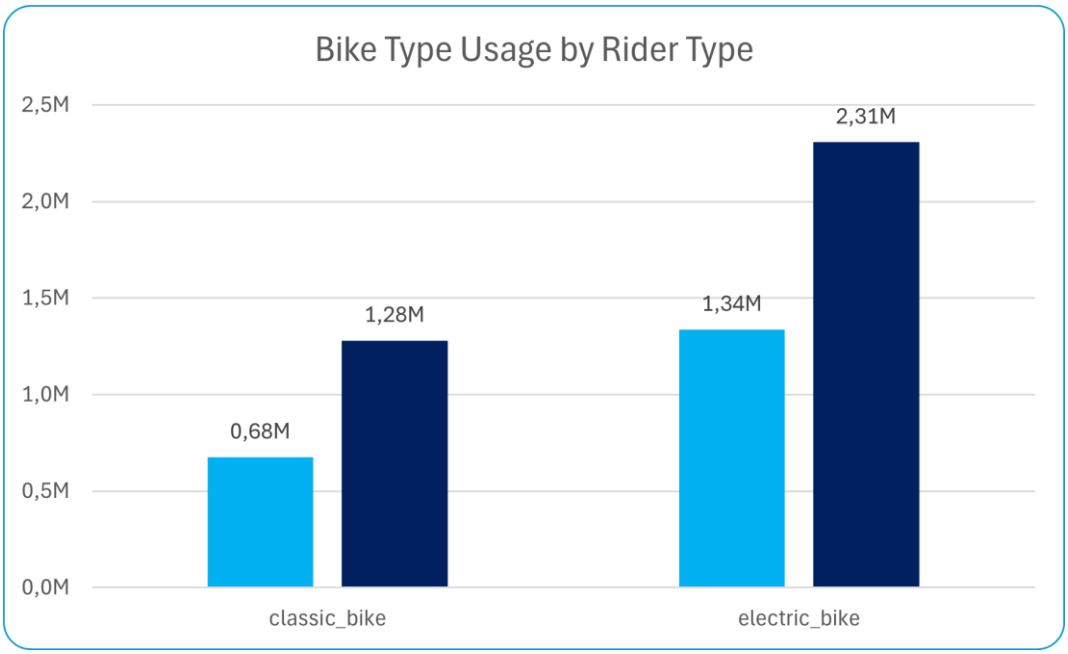

Figure 4. Bike type usage by rider type
Source: Cyclistic dataset (2026), author's analysis.

This insight suggests that electric bikes may represent an opportunity to attract casual riders by emphasizing comfort and accessibility, potentially serving as an entry point toward membership conversion.

Dashboard Design Decisions

Several formatting adjustments were applied to improve interpretability:

- Large numeric values formatted into readable units (e.g., millions).
- Time values converted into minutes.
- Weekdays sorted chronologically using helper columns.
- Labels simplified to avoid visual clutter.

These changes ensured the dashboard communicates insights clearly to non-technical stakeholders.

Based on the analysis, several consistent behavioural differences between casual riders and annual members were identified.

Findings

Based on the analysis, several consistent behavioural differences between casual riders and annual members were identified.

1. Members Ride More Frequently

Annual members generate a higher number of rides overall. This indicates that subscription users rely on Cyclistic bikes as part of their regular mobility habits.

Key insight:
Membership promotes repeated usage and long-term engagement.

2. Casual Riders Take Longer Trips

Casual users show longer average ride durations compared to members.

This suggests that casual riders are more likely using bikes for:

- sightseeing
- leisure activities
- occasional outings

rather than daily transportation.

Key insight:
Casual riders value experience and flexibility more than efficiency.

3. Weekly Usage Patterns Differ Significantly

The weekday analysis reveals one of the strongest behavioural contrasts:

- Members ride consistently during weekdays.
- Casual riders peak during weekends.

This pattern strongly supports the idea that members use bikes primarily for commuting, while casual riders use them recreationally.

Key insight:
User type strongly influences when bikes are used.

4. Opportunity for Conversion

The differences observed suggest that casual riders already see value in the service but interact with it differently.

Because casual riders:

- already use the bikes,
- spend longer time riding,
- and are active during predictable periods,

they represent a strong target group for conversion into annual members.

Transition to Recommendations

The findings highlight clear behavioural distinctions between user types. Understanding these differences provides the foundation for developing marketing strategies aimed at converting casual riders into annual members, which will be explored in the next section.

Recommendations

Based on the analysis conducted throughout this case study, several actionable recommendations can be proposed to help Cyclistic convert casual riders into annual members.

1. Promote Memberships During Weekend Usage

The analysis shows that casual riders are significantly more active during weekends, while members use the service more consistently throughout the week. This suggests that casual riders primarily use bicycles for leisure rather than commuting.

Cyclistic could introduce targeted weekend campaigns promoting membership benefits directly within the mobile app or at docking stations. Offering limited-time discounts after multiple weekend rides could encourage casual users to consider long-term subscriptions.

2. Highlight Cost Savings for Frequent Riders

Although casual riders take fewer trips overall, their average ride duration is longer. This indicates that many casual users may already be paying higher total costs compared to a membership plan.

Cyclistic should clearly communicate potential savings by showing users personalized comparisons such as:

- "You could have saved X this month with a membership."

This type of data-driven messaging can help users recognize the financial advantages of becoming members.

3. Design Marketing Around Leisure-Oriented Experiences

Casual riders appear to use bikes mainly for recreational purposes. Marketing strategies could focus on leisure experiences such as:

- city exploration routes
- tourist partnerships
- weekend activity campaigns

Positioning memberships as a convenient way to enjoy the city regularly may increase conversion rates.

4. Optimize Bike Availability Based on Usage Patterns

Since demand from casual riders increases during weekends and peak leisure hours, Cyclistic could adjust bike distribution strategies to ensure higher availability in popular recreational areas.

Improved availability enhances user experience and increases the likelihood of repeated usage and eventual membership conversion.

5. Encourage First Membership Through Trial Incentives

Introducing short trial memberships or discounted first-month subscriptions could reduce the psychological barrier between casual usage and full membership adoption.

A smooth transition experience may significantly improve conversion rates.

Limitations

Although the analysis provides useful insights, several limitations should be considered:

The dataset does not include demographic or geographic user information, limiting deeper behavioral segmentation.

Weather conditions and seasonal events were not incorporated, which may influence ride patterns.

The analysis focuses on aggregated trends rather than individual user journeys.

Future analyses could integrate additional contextual variables to improve predictive insights.

Conclusion

This case study analysed Cyclistic's bike-sharing data to identify behavioural differences between casual riders and annual members and explore opportunities to increase membership conversions.

The analysis revealed clear distinctions between both user groups. Members tend to use the service more frequently and consistently throughout the week, suggesting commuting or routine-based usage. In contrast, casual riders show higher activity during weekends and longer ride durations, indicating a more leisure-oriented experience.

By leveraging these insights, Cyclistic can implement targeted marketing strategies focused on weekend engagement, cost awareness, and user experience improvements to encourage casual riders to transition into annual members.

Overall, this project demonstrates how data-driven decision-making can support business strategy and customer growth within a bike-sharing environment.